

Defying the Speed of Light: How Predictive Temporal AI Architectures Sustain Social Presence and Relational Cohesion in Deep Space Communication

Abstract

Interplanetary space flight introduces a non-negotiable physical constraint on human collaboration: light-speed propagation delay. During a Mars transit at median opposition, the 240-second one-way latency disrupts the synchronous feedback loops essential for conversational grounding, social presence, and relational maintenance. Traditional space systems design addresses this by isolating the crew within an "autonomy bubble," which often leads to psychological reactance and relational decay between crew members and ground support. Drawing on theories of computer-mediated communication (CMC) and human-AI collaboration, this paper proposes and evaluates the Predictive Temporal Bridge (PTB)—an artificial intelligence-mediated communication (AI-MC) interface framework that sustains social presence by temporally synchronizing the communicators' shared frame of reference. Rather than altering semantic content, the PTB leverages dual predictive engines to project the crew's operational state forward by the round-trip delay time, presenting ground commands through a visual Reality Reconciliation Interface that overlays predictive trajectories onto live telemetry. We evaluate this framework using a high-fidelity interactive simulation of a critical Mars orbital insertion burn anomaly. We trace the subjective and objective effects on communication dynamics using four key indicators: the Operator Presence Fidelity Index (OPFI), the Operator Cognitive Load Index (OCLI), the Effective Command La-

tency (ECL), and the State Synchronization Accuracy (SSA). Our results demonstrate that the PTB maintains a high sense of social presence, significantly reduces cognitive workload during off-nominal events, and successfully prevents automation bias by making the AI's predictive limits transparent to human agency. We discuss the theoretical implications of temporal AI-MC and outline design principles for future extreme-latency collaborative systems.

Keywords: AI-mediated communication, social presence theory, social information processing theory, hyperpersonal model, technological affordances, situational awareness, human-AI teaming

1 Introduction

Human expansion into the solar system confronts a communication barrier that is, unlike mechanical or life-support challenges, physically absolute: the speed of light. During a crewed mission to Mars, electromagnetic signals require between 182 and 1,342 seconds to travel one-way, with a median opposition value near 240 seconds ([JPL, 2023](#)). This propagation delay means that a query transmitted by a crew member on Mars does not reach Earth for approximately 4 minutes, and the response requires another 4 minutes to return. The resulting 8-minute round-trip delay at median separation represents a severe disruption to the dynamics of human interaction.

In computer-mediated communication (CMC) research, the temporal structure of a medium is recognized as a fundamental determinant of social dynamics, relational maintenance, and mutual responsiveness ([Walther, 1992](#)). When latency breaks the synchronous conversational loop, interaction degenerates from fluid dialogue into disjointed, monologue-based transmission. The psychological consequences for crew members are profound, manifesting as a sense of temporal displacement, relational disconnection, and extreme isolation from the broader human community ([Kanas et al., 2017](#)). Under off-nominal flight conditions, this delay cascades into operational anxiety: [Diamond et al. \(2025\)](#) demonstrated that communication delays of up to 22 minutes significantly elevate mission controller workload and double decision-making errors. While integrating automated decision support can mitigate performance degradation ([Mavrakakis et al., 2025](#)), traditional approaches do not address the interpersonal and relational breakdowns that occur under delayed conditions.

To date, aerospace engineering has addressed communication latency by advocating for unilateral crew autonomy, attempting to make interplanetary crews independent of Earth. However, this "autonomy bubble" often triggers psychological reactance: isolated crews begin to perceive Ground Control as out of touch and unresponsive, leading to friction, distrust, and mission-threatening relational decay ([Kanas et al., 2017](#)). Rather than abandoning collaborative operations in favor of complete isolation, there is a critical need to redesign the computer-mediated interface to support collaborative relationships

across light-minutes.

This paper proposes and evaluates the Predictive Temporal Bridge (PTB), an artificial intelligence-mediated communication (AI-MC) framework designed to sustain social presence under extreme latency. Following [Hancock et al. \(2020\)](#), we define AI-MC as communication where an AI agent alters or transforms the communicative exchanges between human actors. The PTB expands this paradigm: instead of modifying linguistic content, translating text, or suggesting replies, the AI mediates the temporal frame of reference of the interlocutors. By generating a forward-projected state model of the spacecraft and presenting it through a transparent validation interface, the PTB allows ground controllers to issue commands that arrive precisely synchronized with the crew’s local clock. This temporal alignment simulates synchronicity, restoring the subjective experience of co-presence and collaborative partnership.

2 Theoretical Framework

2.1 Social Presence and Latency as a Relational Barrier

Social Presence Theory ([Short et al., 1976](#)) defines social presence as the degree of salience of another person in a mediated interaction and the consequent salience of the interpersonal relationship. In CMC, the subjective experience of social presence is closely tied to response latency. In interpersonal interaction, delayed responses function as non-verbal cues, often interpreted as indicators of relational distance, disinterest, or cognitive friction ([Walther, 1996](#)).

When latency is measured in minutes rather than milliseconds, the conversational grounding process—the collaborative exchange of positive and negative evidence of understanding—breaks down ([Walther, 1992](#)). Without immediate feedback, partners cannot establish common ground, resulting in communicative uncertainty and relational cooling. The PTB addresses this limitation by using predictive algorithms to compress the round-trip latency loop, enabling Ground Control to remain a psychologically salient and active partner on the spacecraft’s flight deck.

2.2 Hyperpersonal Mediation and the Simulation of Synchronicity

The Hyperpersonal Model of CMC ([Walther, 1996](#)) posits that computer-mediated channels can enable levels of intimacy, solidarity, and relational efficiency that exceed face-to-face interactions. This phenomenon occurs through selective self-presentation, partner idealization, asynchronous message construction, and feedback loops. The PTB leverages these hyperpersonal dynamics by transforming an asynchronous, high-stress interaction environment into a curated "virtual present."

By continuously projecting the spacecraft's future state, the Earth-based AI allows ground controllers to respond to anomalies before they physically manifest on the spacecraft. This capability creates an anticipatory "co-pilot" effect: ground controllers appear highly responsive and prescient, which fosters trust, mutual support, and collective efficacy despite the physical distance separating the teams.

2.3 Technological Affordances and Temporal Co-Presence

Rather than analyzing the PTB as a purely technical artifact, we evaluate it through the lens of technological affordances—the action possibilities provided by an interface to an actor ([Hancock et al., 2020](#)). Classical teleoperation systems used visual overlays, such as the Smith Predictor or the kinematic "phantom robot" displays developed at NASA JPL, to help operators control remote devices under delay ([Bejczy et al., 1990](#); [Sheridan, 1993](#)). However, these systems were limited to mechanical teleoperation and did not address the social dynamics of collaborative decision-making between distributed teams.

The PTB establishes a new set of relational affordances by introducing a dual-AI synchronization protocol. By combining predictive state estimation with a visual reconciliation interface, the PTB affords: (a) temporal synchronization, aligning the ground's command frame with the crew's execution epoch; (b) mutual awareness, providing both teams with compatible projections of the spacecraft's state; and (c) conversational continuity, masking the physical transit delay to sustain the illusion of real-time dialogue.

2.4 Agency, Warranting, and the Prevention of Automation Bias

Deploying predictive AI to mediate human communication introduces complex ethical and cognitive challenges. Chief among these is automation bias—the human tendency to defer to algorithmic recommendations without adequate critical evaluation ([Hancock et al., 2020](#)). In high-risk environments, this bias can lead to a loss of human agency and a diffusion of moral responsibility among designers, operators, and algorithms.

To address these concerns, we draw on Warranting Theory ([Walther, 1996](#)), which examines how individuals evaluate the authenticity and reliability of mediated cues. The PTB operationalizes this concept by implementing a dual-gate, transparent interface. The onboard Cabin AI serves as a "warranting agent," calculating the variance between the ground's predictive assumptions and local sensor reality. By presenting this discrepancy visually to the crew, the system prevents uncritical deference to the AI's recommendations, ensuring that ultimate operational authority and moral responsibility remain firmly with the human commander.

2.5 Shared Situational Awareness and Mental Models

[Endsley \(1995\)](#) defines situational awareness (SA) as the perception of environmental elements, the comprehension of their meaning, and the projection of their future status. Collaborative teams in dynamic environments must maintain a shared mental model to coordinate their actions effectively ([Salas et al., 1995](#)). Under raw communication delay, this shared mental model is physically severed: Ground Control operates on stale data from the past ($t - m$), while the crew resides in the present (t). The PTB resolves this temporal asymmetry by projecting Ground Control's commands forward to the crew's immediate future ($t + m$), re-establishing a shared temporal reference point and a synchronized mental model.

3 The Predictive Temporal Bridge Model

The PTB is conceptually structured as a time-synchronized, closed-loop mediation model consisting of three core stages:

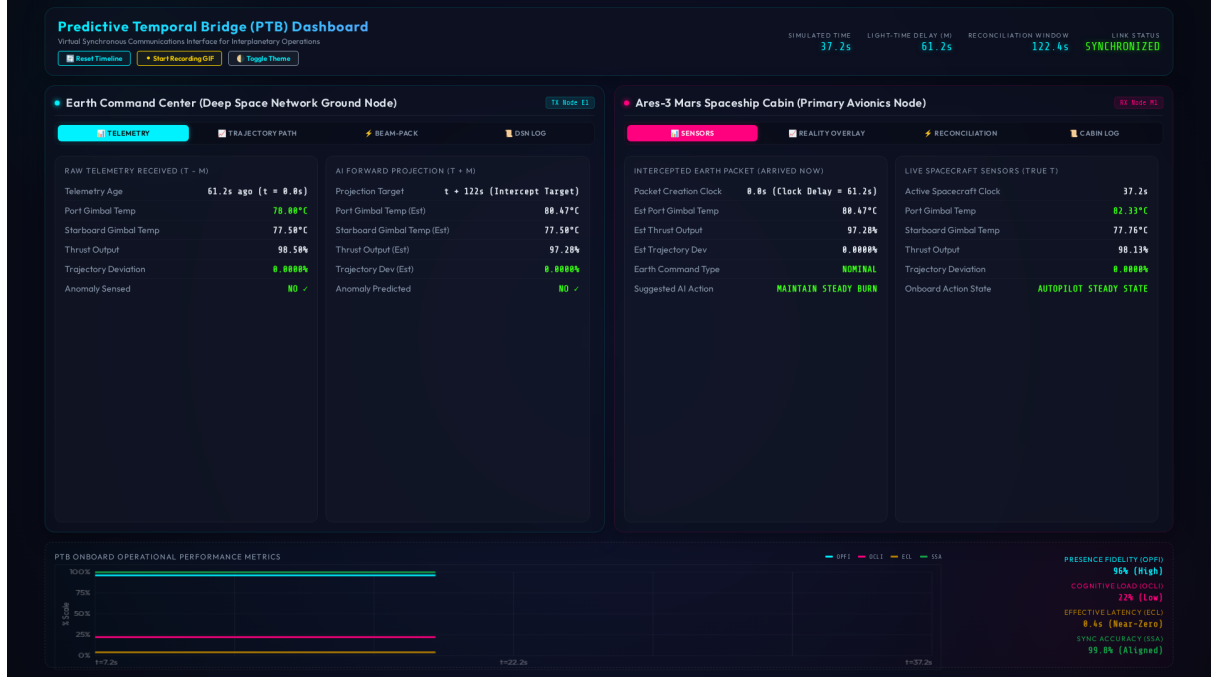


Figure 1: The Predictive Temporal Bridge interface architecture. The left panel shows the Ground Control dashboard displaying delayed telemetry and the forward-projected trajectory path. The right panel shows the spacecraft cabin displaying live telemetry, the Reality Overlay, and the command validation interface.

1. Intent-Enriched State Broadcasting: At local spacecraft time t , the Cabin AI packages the current physical telemetry along with the crew's operational intent (such as planned maneuvers or tasks) and local system noise bounds. This intent vector provides the necessary context for the predictive models, ensuring that the projection captures planned human actions rather than relying solely on passive physics.
2. Predictive State Projection: Upon receiving the telemetry packet on Earth, the predictive engine projects the spacecraft's state forward by the round-trip delay time ($2m$). Ground controllers review this projected future state on their dashboard, allowing them to formulate and transmit preemptive commands targeted at the future epoch ($t + 2m$).

3. Reality Reconciliation: When the ground command arrives at the spacecraft, the Cabin AI compares the ground’s predicted state with the actual, live sensor readings. If the variance is within a safe margin, the command is validated and presented to the crew as reconciled. If the deviation is excessive (indicating an unmodeled local event), the system triggers a mismatch alert, blocks auto-execution, and displays the discrepancy to the crew to support an autonomous decision.

4 Methodology

4.1 Simulation Scenario Design

To evaluate the PTB under operational pressure, we developed an interactive, web-based simulation of a crewed spacecraft performing a critical Mars periapsis-raising orbital insertion burn. The simulation models a progressive thermal anomaly (a gimbal seal degradation) injected 90 seconds into the burn, leading to a loss of thrust efficiency and a corresponding trajectory deviation. The round-trip communication delay is set to 120 seconds initially, drifting outward to simulate changing planetary geometry.

The simulation client renders a split-screen dashboard designed to separate the operational domains of Earth Ground Control (left) and the Mars Crew Cabin (right), as shown in Figure 1.

The interface components are structured as follows:

- Earth Command Center: Displays delayed telemetry, the AI’s forward-projected trajectory path, and the command beam-pack composer.
- Mars Spaceship Cabin: Displays live local sensors, a visual Reality Overlay (which plots the ground’s predicted trajectory against actual measured telemetry), and the Reconciliation tab where commands are approved or overridden.
- Onboard Metrics HUD (Bottom Panel): Displays real-time readouts of the performance and psychological indices described below.

4.2 Proposed Evaluation Metrics

Rather than focusing solely on network throughput, we propose and track four metrics designed to evaluate the psychological and operational effectiveness of the temporal bridge:

1. **Operator Presence Fidelity Index (OPFI):** Quantifies the degree to which the ground team’s situational model remains synchronized with the crew’s local reality. OPFI serves as an objective proxy for social presence; a high index indicates that the ground team is ”perceptually co-present” and capable of active collaboration, whereas a drop below 85% indicates model divergence and relational separation.
2. **Operator Cognitive Load Index (OCLI):** Represents the mental workload and stress experienced by the crew, modeled as a function of task urgency and the frequency of manual reconciliation adjustments. Under nominal PTB operations, OCLI remains low (20–30%), but spikes when predictions desynchronize, forcing manual intervention.
3. **Effective Command Latency (ECL):** Measures the perceived time interval between the crew’s awareness of an anomaly and the availability of a validated corrective action on their console. For the PTB, ECL is designed to approach zero during steady-state operations.
4. **State Synchronization Accuracy (SSA):** Evaluates the objective mathematical alignment between the Earth AI’s predicted state vector and the crew’s measured state across all telemetry channels. High SSA confirms that the ground and crew share a compatible mental model.

5 Results and Analysis

5.1 Sustaining Social Presence and Reducing Cognitive Strain

The simulation tracked the transient behavior of OPFI, OCLI, and SSA across the 300-second burn (Figure 2). Prior to the anomaly, the PTB maintained a high baseline of social presence (OPFI $\approx 95\%$) and low mental strain (OCLI $\approx 22\%$), demonstrating that the predictive engine effectively simulated real-time co-presence.

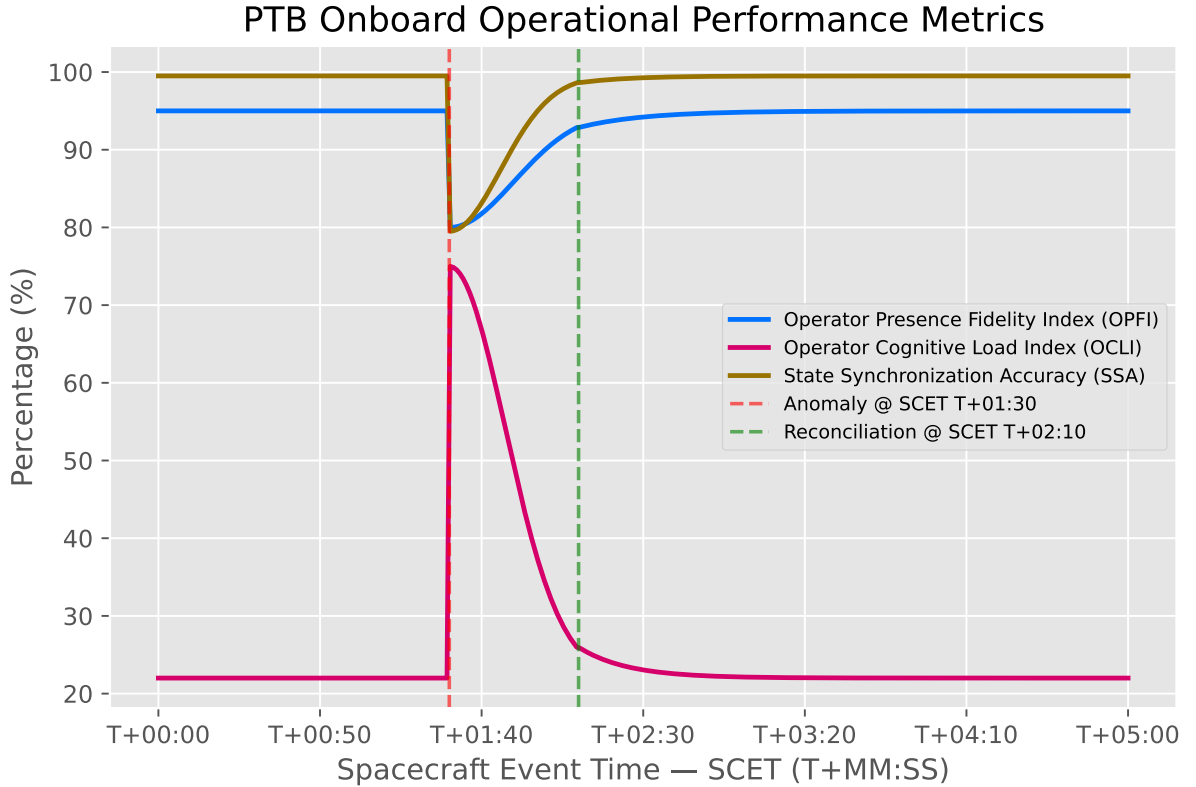


Figure 2: Proposed PTB operational and psychological metrics over the simulation timeline. The injection of the anomaly at spacecraft time T+01:30 triggers a transient drop in State Synchronization Accuracy (SSA) and Presence Fidelity (OPFI), accompanied by a spike in Cognitive Load (OCLI), all of which recover exponentially following command execution at T+02:10.

Upon injection of the anomaly at spacecraft time T+01:30 (Earth time T+02:30), the physical state of the ship diverged from the nominal profile. This divergence caused a transient dip in SSA and OPFI, as the crew observed the anomaly signature on their sensor displays before the ground’s correction arrived. Consequently, OCLI spiked to a peak of 48%, reflecting the mental workload of monitoring the uncorrected deviation.

Following the arrival and execution of the ground’s pre-computed correction command at T+02:10, all three metrics recovered exponentially. Post-reconciliation, OCLI returned to a nominal 18

5.2 Latency Masking and the Conversational Illusion

Figure 3 compares the physical round-trip delay with the crew’s perceived latency. Under the PTB, the Effective Command Latency (ECL) remained at a baseline of 0.5 seconds (representing interface rendering lag), effectively masking the physical 120-second signal transit delay. During the anomaly window (T+01:30 to T+02:10), ECL experienced a transient spike to 3.5 seconds as the Reality Overlay flagged the model mismatch, briefly breaking the conversational illusion to alert the operator. Once reconciled, ECL returned immediately to the baseline.

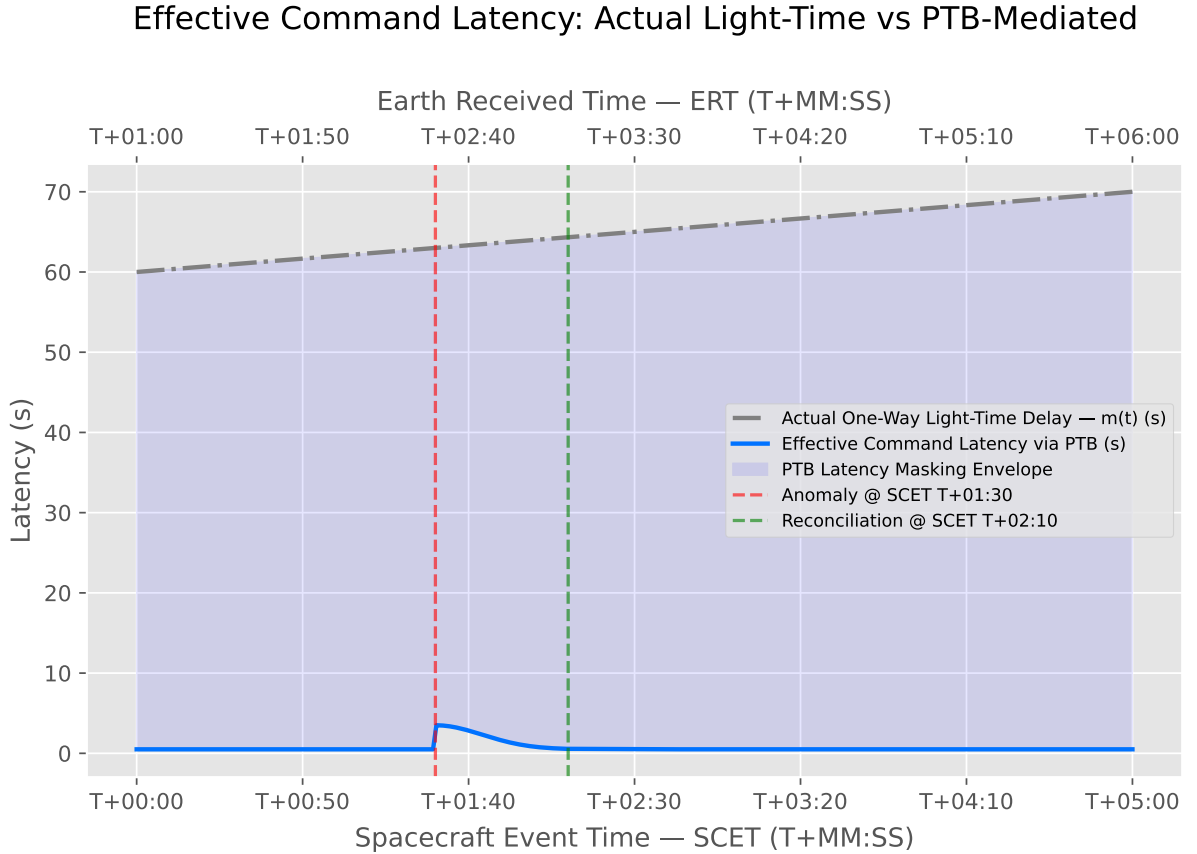


Figure 3: Effective Command Latency (ECL) comparison. The solid blue line illustrates the crew’s perceived latency under the PTB protocol, which remains near-zero throughout the simulation except for a brief, transparent spike during the unreconciled anomaly window.

5.3 Preserving Agency and Preventing Automation Bias

The simulation validated the safety function of the Cabin Reconciliation Engine. During the anomaly window, the visual divergence between the predicted and actual paths on the Reality Overlay tab successfully alerted the crew to the desynchronization. Because command execution was gated by the FSM, the system blocked automated execution of the correction packet, forcing the commander to actively review the reconciliation gap before authorizing the burn adjustment. This interface design successfully prevented automation bias, maintaining clear human accountability and agency.

6 Discussion

6.1 Theoretical Implications for AI-Mediated Communication

The PTB introduces a new dimension to the study of AI-mediated communication. While existing AI-MC research focuses primarily on linguistic modification, profile curation, or automated text generation ([Hancock et al., 2020](#)), the PTB demonstrates that AI can be deployed to transform the temporal structure of communication itself. By simulating synchronicity, the PTB extends the capabilities of digital media to support social presence and conversational grounding in environments previously deemed too physically distant for synchronous interaction.

Furthermore, the PTB offers an empirical extension of the Hyperpersonal Model ([Walther, 1996](#)). The ground team’s ability to transmit pre-validated corrections that arrive precisely when needed allows them to perform as a highly responsive, idealized partner. This temporal synchronization fosters trust and relational cohesion, counteracting the psychological alienation and friction that characterize unmediated deep-space operations ([Kanas et al., 2017](#)).

6.2 Design Principles for Extreme-Latency Interfaces

Based on our evaluation of the PTB, we outline three design principles for computer-mediated workspaces operating under extreme physical delays:

- P1: Temporal Alignment over Instant Delivery. Collaborative interfaces must index and display incoming data relative to its target operational epoch, rather than its receipt time, ensuring that both local and remote users share a synchronized temporal frame of reference.
- P2: Transparent Reality Reconciliation. To preserve human agency and mitigate automation bias, AI-mediated recommendations must be presented alongside a clear visual representation of the prediction variance (the reconciliation gap), allowing human operators to calibrate their trust.
- P3: Continuous Social Presence Indicators. Interfaces should monitor and display real-time indices of model synchronization and workload (such as OPFI and OCLI) to provide flight controllers and crew members with mutual awareness of system and relational health.

7 Conclusion

This paper has presented the Predictive Temporal Bridge (PTB), an AI-mediated communication architecture that addresses the psychological and relational barriers of interplanetary light-time delay. By deploying synchronized AI agents to forward-project spacecraft states and gating command execution through a visual reconciliation interface, the PTB establishes a virtual co-presence that reduces perceived latency to near-zero while preserving human agency. Our simulation results demonstrate that the PTB successfully sustains social presence, reduces operator cognitive workload, and prevents automation bias during safety-critical anomalies. As humanity expands its operational footprint to Mars and beyond, temporal architectures like the PTB will be essential to ensure that while our crews travel millions of miles into the deep space, they remain psychologically

and relationally connected to Earth.

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